

CVaR / Expected Shortfall

VaR tells you where the tail begins. Expected shortfall tells you what lives inside it — which is where portfolios actually die.

Value-at-Risk tells you where the tail begins. It says nothing about what lives inside it — and inside the tail is where portfolios actually die. Two books can report identical 99% VaR yet differ threefold in what they lose once the threshold breaks; for fat-tailed assets like high-yield credit, that difference is the whole risk. Expected shortfall (CVaR) fixes the blind spot by averaging the tail instead of pointing at its door, which is why Basel made it the regulatory standard. We estimate both, historically and with a Student-t, on 4,462 days of HYG and VWO — a sample that deliberately includes the crisis these measures were built for.

PROJECT CARD

CATEGORY

Risk Management

LIBRARIES

NumPy · SciPy · Pandas · Riskfolio-Lib

ASSETS

HYG + VWO

TIMEFRAME

Apr 2007 – Dec 2024

USE

Setting tail-risk limits that survive a crisis — and pass a regulator

- 01 Load HYG + VWO daily closes, 2007–2024 — the GFC included
- 02 Build two synthetic books with identical 99% VaR, very different tails

- 03 Estimate 97.5% / 99% VaR & CVaR — historical and Student-t (Acerbi's formula)
- 04 Test subadditivity: CVaR always passes, VaR fails in crisis-year samples

05 Zoom into HYG's GFC tail against its full-sample 99% VaR

06 Spend the same 4% tail budget through $rm="MV"$ vs $rm="CVaR"$ (Riskfolio-Lib)

1. What VaR cannot see

VaR at confidence α is a **quantile**: the smallest loss exceeded on only the worst $(1-\alpha)$ of days. Read that definition again and notice what it does not say — nothing about **how much** you lose on those days. VaR is a threshold, not an average, so it is structurally blind to everything beyond itself. A ten-line thought experiment makes the blindness concrete: build two 1,000-day P&L histories that agree on 990 benign days **and** on the day that sets the 99% quantile, but whose ten worst days differ by a factor of six.

WHY PRACTITIONERS CARE

Basel's Fundamental Review of the Trading Book (FRTB) replaced 99% VaR with 97.5% expected shortfall as the capital measure for trading desks — precisely because VaR ignores tail severity and can punish diversification. The 97.5% level was chosen so that, for a normal distribution, ES roughly matches the old 99% VaR — but unlike VaR it keeps paying attention when the tail turns out fatter than normal.

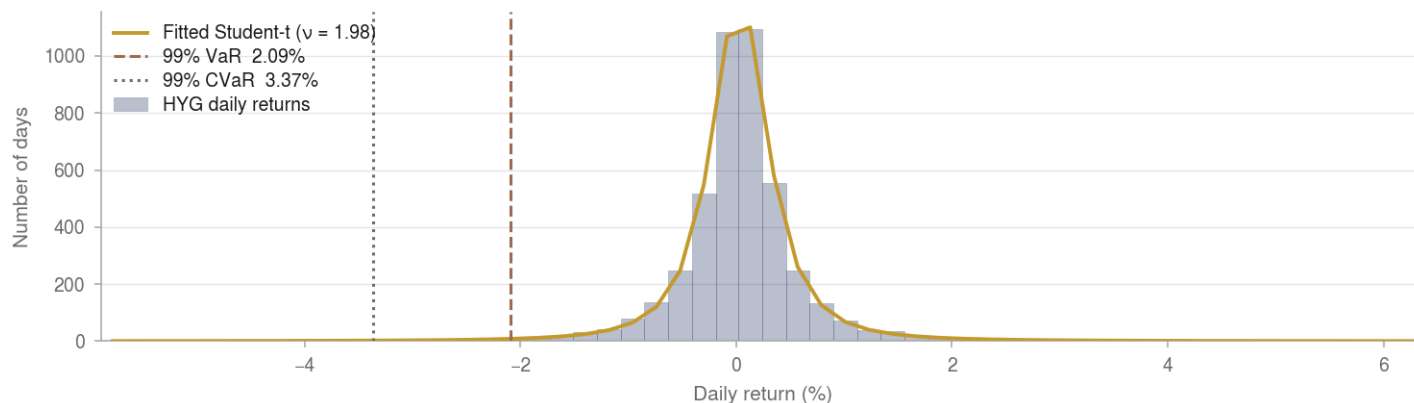
Both books report a 99% VaR of 2.40%, so a risk report built on VaR alone calls them equally risky. Yet book A's CVaR is 2.60% against book B's 7.26%, with worst days of -2.60% and -15.00% — identical VaR, 2.8× the expected tail loss. CVaR — the **mean** loss conditional on breaching the VaR quantile — separates the two books instantly, because it averages over the whole tail instead of reading one point at its edge.

2. Two fat-tailed assets

Now the real thing. HYG (iShares high-yield corporate bond ETF) and VWO (Vanguard emerging-markets equity) are chosen because they are risky in opposite ways. EM equity is honestly volatile — it looks risky and it is. Credit is the trap: it collects small, steady coupons in calm markets and hands them back all at once in a crisis. On paper HYG's 11.2% annualised vol looks far safer than VWO's 27.1% — until you notice that the quiet series hides a worst day of -8.10% . The aligned sample runs 2007-04-11 (HYG's listing) to 2024-12-31. A Student-t fitted to HYG's daily returns lands at $\nu \approx 1.98$ degrees of freedom — below 2, which means the fitted distribution does not even possess a finite variance. Read that as a diagnostic as much as an estimate: an unconditional iid fit has nowhere to put 2008's volatility clustering except the tail parameter, so it buys realism at the extremes by overstating how wild a **typical** day is. (Condition on a GARCH filter and the residual df comes out higher; the unconditional fit is the honest worst case.)

HYG daily returns 2007 → 2024 with fitted Student-t ($\nu = 1.98$) — VaR marks the door, CVaR the room behind it

The horizontal gap between the two thresholds is everything VaR does not price



Source: Author calculations, from the article's computed results.

The horizontal gap between the two dashed lines — 2.09% to 3.37% — is everything VaR does not price, and HYG's gap is the widest of the three books we measure.

3. Historical and Student-t estimates

Estimating CVaR comes down to two philosophies. The historical estimator trusts the data: sort, take the quantile, average everything beyond it. The parametric route trusts the model: fit a Student-t and use **Acerbi's closed-form expected shortfall** — the analytic mean of the t's tail — which can extrapolate to days worse than anything the sample has actually seen:

```
def var_hist(x, a):
    return -np.quantile(x, 1 - a)

def cvar_hist(x, a):
    q = np.quantile(x, 1 - a)
    return -x[x <= q].mean()

def cvar_t(params, a):
    # Acerbi & Tasche closed form
    nu, loc, scale = params
    p = 1 - a
    xp = stats.t.ppf(p, nu)
    return -loc + scale * stats.t.pdf(xp, nu) * (nu + xp**2) / ((nu - 1) * p)
```

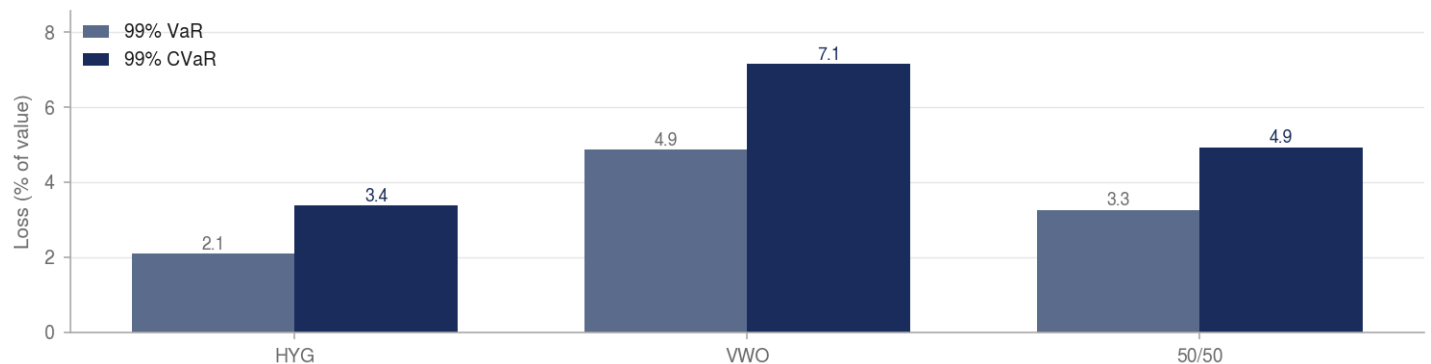
HYG estimate	97.5%	99%
Historical VaR	1.28%	2.09%
Historical CVaR	2.29%	3.37%
Student-t VaR ($\nu = 1.98$)	1.21%	1.99%
Student-t CVaR (Acerbi)	2.54%	4.09%

Three things to read off that table:

- **The FRTB calibration at work** — HYG's 97.5% CVaR (2.29%) sits close to its 99% VaR (2.09%): similar magnitude, but the CVaR number keeps growing when the tail does.
- **The t extrapolates past the sample** — its CVaR (4.09%) exceeds the historical one because with $\nu \approx 1.98$ the fitted tail expects days worse than any yet observed.
- **The small-sample caveat** — at 99% the historical CVaR is the mean of just 45 observations. The estimator with the best theoretical properties stands on the fewest data points, which is the practical argument for fitting a parametric tail and letting it extrapolate.

99% VaR vs 99% CVaR, historical — per asset and for the daily-rebalanced 50/50

For every book the CVaR bar runs roughly one and a half times the VaR bar



Source: Author calculations, from the article's computed results.

Asset	Ann. vol	99% VaR	99% CVaR	CVaR/VaR	Worst day
HYG	11.2%	2.09%	3.37%	1.62×	-8.10% (2008-09-29)
VWO	27.1%	4.86%	7.15%	1.47×	-15.34% (2008-10-15)
50/50	17.5%	3.26%	4.92%	1.51×	-9.85% (2008-10-15)

Read the first and fifth columns together. HYG's volatility is well under half of VWO's, yet its CVaR/VaR ratio of 1.62× is the **highest** in the table — credit's deceptively quiet variance hides the most disproportionate tail. Any tool that ranks these assets by σ alone has the risk ordering half wrong.

4. Subadditivity — the coherence test

In 1999, Artzner, Delbaen, Eber and Heath wrote down four axioms any sane risk measure should satisfy. VaR fails one — and it is the one your intuition cares most about: **subadditivity**, $\rho(A+B) \leq \rho(A) + \rho(B)$, the rule that diversification must never create risk. The classic counterexample needs only two independent bonds, each defaulting with probability 0.7%. Held alone, each has **zero** 99% VaR: a 0.7% loss probability hides entirely below the 1% threshold, so VaR literally cannot see it. Mix them 50/50 and the book loses on 1.4% of scenarios — above 1% — so the diversified book has strictly **positive** 99% VaR. Diversifying “created” risk,

says VaR. The risk was there all along, of course. VaR was hiding it.

It is not just a parlour trick. On the full HYG/VWO sample the 50/50 portfolio behaves — 99% VaR of 3.26% against a weighted blend of 3.47%. But scan calendar-year sub-samples across confidence levels and historical VaR breaks subadditivity 64 times; the worst case is 2009 at $\alpha = 98.8\%$, where the portfolio's VaR of 4.48% exceeds the blend's 3.95%. CVaR, by construction — Rockafellar & Uryasev's convex formulation makes this explicit — never violates: 4.92% for the portfolio versus 5.26% for the blend at 99%, and it passes at every year and level where VaR fails.

NOTE

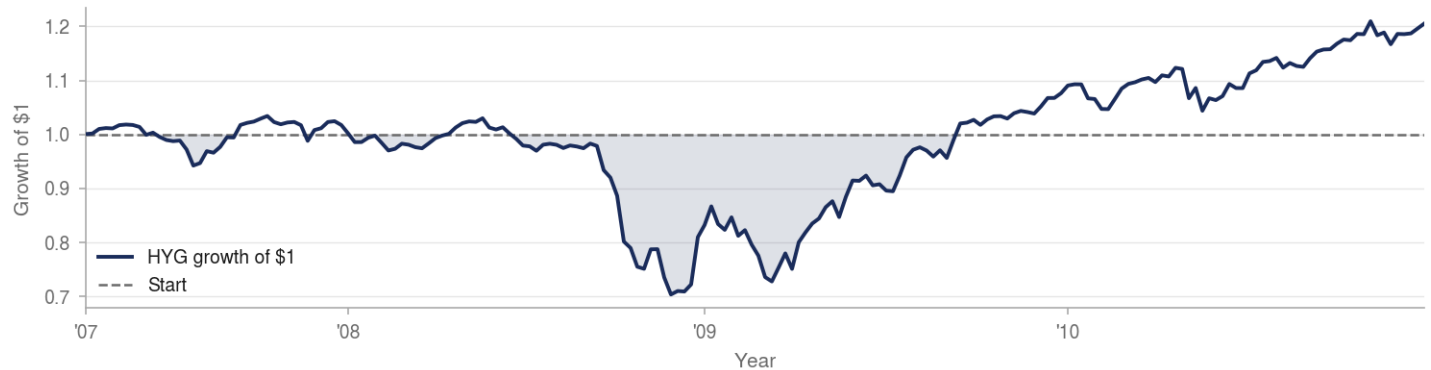
Subadditivity is what makes a risk measure **aggregable**: desk CVaRs can be summed into a conservative firm-wide bound, and CVaR limits can be pushed through a convex optimiser. Neither is true of VaR — a VaR-constrained optimisation is non-convex and can reward concentration.

5. The GFC, seen from the tail

HYG's full-sample 99% VaR is 2.09%. Now watch the crisis ignore it: from September 2008 through March 2009, HYG breached that threshold on 24 of 146 trading days — a 1% tail arriving at 24× its expected frequency of ~1.5 days. The worst prints came fast: -8.10% on 2008-09-29, -6.67% on 2008-10-10, -6.01% on 2008-09-17 — every one of them multiples of the VaR line, in an ETF marketed as a bond fund.

HYG cumulative return, 2007-04-11 → 2010 — the GFC drawdown that a 2.09% VaR never described

Roughly flat, then a collapse to about a third below its start into early 2009, then recovery



Source: Author calculations, from the article's computed results.

At the trough a dollar invested at HYG's listing was worth \$0.69. VaR answered "how often?" — and even that answer failed under regime change. CVaR at least asks the question that mattered in 2008: **how bad is it when it happens?**

6. Spending a tail budget: CVaR vs MV

So far CVaR has been the better thermometer. The sharper question is whether it changes what you **do**. With two assets, pure risk-minimising is degenerate here — volatility and tail agree that HYG is the quieter asset, so both $rm="MV"$ and $rm="CVaR"$ corner at 100% HYG. The measures diverge the moment you **spend a risk budget**. Hand two desks the same mandate — an expected loss on the worst 1% of days of at most 4% — and let each maximise return against it. The MV desk speaks only volatility, so it must translate the budget through a normal distribution ($ES = 2.665\sigma$ for a Gaussian, so $\sigma \leq 1.50\%$ daily); the CVaR desk constrains the realised tail directly:

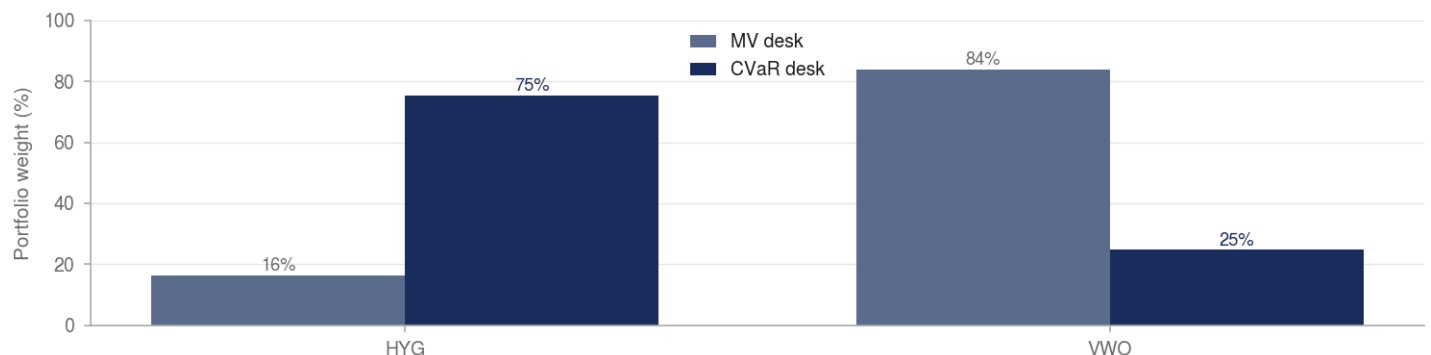
```
import riskfolio as rp

p_mv = rp.Portfolio(returns=rets[["HYG", "VW0"]])
p_mv.assets_stats(method_mu="hist", method_cov="hist")
p_mv.upperdev = 0.015 # 4% ES budget, normal-translated
w_mv = p_mv.optimization(model="Classic", rm="MV", obj="MaxRet", hist=True)

p_cv = rp.Portfolio(returns=rets[["HYG", "VW0"]])
p_cv.assets_stats(method_mu="hist", method_cov="hist")
p_cv.alpha = 0.01
p_cv.upperCVaR = 0.04 # the tail budget itself
w_cv = p_cv.optimization(model="Classic", rm="CVaR", obj="MaxRet", hist=True)
```

Same 4% tail budget, two allocations — how each desk spends it

The MV desk puts 83.8% in VW0 where the CVaR desk allows only 24.7%



Source: Author calculations, from the article's computed results.

Desk	Ann. return	Ann. vol	Realised 99% CVaR
MV (vol-translated budget)	6.6%	23.8%	6.37% — budget was 4%
CVaR (direct)	5.7%	13.6%	3.99%

Same stated risk appetite, radically different books: 83.8% VWO through the variance lens versus 24.7% through the CVaR lens. Because both assets' tails are fatter than the Gaussian used in the translation, the MV desk's realised 99% CVaR of 6.37% overshoots its own 4% mandate by roughly 59% — the tail it was told to cap is exactly what its risk measure could not see.

PRACTITIONER TAKE

Use CVaR where the loss distribution is asymmetric or fat-tailed — credit, options, carry, anything with a “smile now, cry later” profile. HYG is the canonical trap: its vol whispers safety while its tail shouts the opposite. Practically: quote 97.5% ES next to any 99% VaR (FRTB's own calibration), fit a Student-t rather than trusting the empirical tail alone when v comes out below ~ 4 , and put the CVaR constraint **inside** the optimiser — Rockafellar–Uryasev makes it a linear program, so there is no computational excuse. The one honest cost of the switch: ES is harder to **backtest** than VaR. A quantile is directly falsifiable by counting breaches; an expected shortfall is not elicitable on its own, only jointly with its VaR — which is why desk validation (Acerbi–Székely) and FRTB's own backtesting still run on VaR exceptions at two confidence levels, even though the capital number is ES.

REFERENCES

1. Artzner, P., Delbaen, F., Eber, J.-M. & Heath, D. (1999). Coherent Measures of Risk. *Mathematical Finance* 9(3), 203–228.
2. Acerbi, C. & Tasche, D. (2002). On the coherence of expected shortfall. *Journal of Banking & Finance* 26(7), 1487–1503.
3. Rockafellar, R.T. & Uryasev, S. (2000). Optimization of Conditional Value-at-Risk. *Journal of Risk* 2(3), 21–41.
4. Acerbi, C. & Székely, B. (2014). Back-testing Expected Shortfall. *Risk Magazine*, December 2014.
5. Basel Committee on Banking Supervision (2019). Minimum capital requirements for market risk (FRTB). Bank for International Settlements.
6. Companion notebook: `cvar-expected-shortfall.ipynb` — reproduces every figure from raw data (fully deterministic; no simulation).